

Rithesh Kumar

516 4th Avenue, San Francisco, CA, USA

☎ (+1)831-266-7201 | ✉ rithhem20@gmail.com | 🌐 rithesh17.github.io | 🐙 Rithesh17 | 🌐 rithesh-kumar-91332a156

Summary

Research Engineer specializing in model-serving and inference platforms, semantic data and retrieval layers, and the deployment pipelines around them. At Accretional, a founding engineer owning the AI inference stack (gRPC embedding + vLLM serving) and the platform's storage, search, and hosting backend. Previously three years at Goldman Sachs running real-time ML systems at scale (1M+ requests/month, 99.9% uptime). MS in Computer Science (UC Santa Cruz).

Work Experience

Founding Engineer - Infrastructure & Platform

Aug. 2025 - Present

Accretional

San Francisco, CA

- Designed and operate the platform's ML inference layer: a production gRPC embedding service (protobuf-aware, embedding arbitrary typed messages via schema descriptors) and a vLLM-based LLM serving manager that spins up named model servers on demand and routes OpenAI-compatible streaming completions, deployed to GPU on Docker and Modal.
- Led the storage and semantic-search layer (Collector), a typed-protobuf data store with hybrid vector and full-text retrieval; integrated sqlite-vec with a hybrid driver model to assist in code search.
- A major contributor for the platform's security and identity layer: a verification service for sensitive operations (hCaptcha, signed time-limited tokens, per-deploy rotating secrets) and Envoy config that propagates user identity into the container runtime, plus per-user request isolation on the Git transport.
- Designed a distributed gRPC service mesh with type-safe protobuf registry and dynamic dispatch, enabling AI agents to route requests and retrieve context across a multi-service inference platform

AI Engineering Intern

Jun. 2024 - Aug. 2024

Accretional

San Francisco, CA

- Built the retrieval layer behind Brilliant's code generation so the assistant writes against a project's actual APIs; tuned a hybrid FAISS index (Flat-L2 + HNSW) for precision and recall on code search.
- Studied open-source LLM internals (Mistral, LLaMA) and applied LoRA fine-tuning for code generation; re-architected the prompt/preset system and built the one-click deploy flow to AWS Lambda and Google Cloud Functions.

Machine Learning Scientist

Jul. 2020 - Jul. 2023

Goldman Sachs

Bengaluru, India

- Developed LSTM models for loan repayment forecasting and optimized inference pipelines through model quantization and caching, achieving 15% accuracy improvement and reducing latency by 60% from 200ms to 80ms
- Led real-time loan processing pipeline by implementing caching and circuit breakers, handling 1M+ requests per month with 99.9% uptime and graceful degradation under load
- Built A/B testing framework and introduced CI/CD processes across ML team, enabling data-driven model rollouts that reduced deployment risk by 80% and cut deployment time by 70%
- Built simulation models to assess COVID-19 impact on lending risk, enabling data-driven decisions during unprecedented scenarios when historical patterns were unreliable

Machine Learning Research Intern

May 2019 - Jul. 2019

Sprinklr

Gurgaon, India

- Built sentiment analysis pipelines for social media content by processing multilingual and fragmented text, achieving 85% accuracy on noisy real-world data
- Applied model optimization techniques including pruning and quantization, reducing model size by 70% while maintaining 95% of original accuracy for production deployment

Education

Master of Science in Computer Science

Sep. 2023 - Jun. 2025

University of California, Santa Cruz

Santa Cruz, CA

- GPA: 3.9/4.0
- Focused on machine learning and AI, particularly medical imaging applications with emphasis on clinical trust and explainability
- Research in CT scan denoising and AI explainability, developing methods that improve image quality by 30% while maintaining diagnostic accuracy
- Explored attention mechanisms and visualization techniques, creating interpretable models that increase confidence by 60%

- GPA: 9.4/10.0, Graduated with Honors
- Focused on ML and AI, with research in medical image analysis resulting in published work on automated cancer grading
- Published work on automated prostate cancer grading achieving 92.38% accuracy, contributing to Springer proceedings
- Collaborated with pathologists to identify key diagnostic features, designing models that achieved 85% agreement with expert assessments

Open Source Projects

Formal Grammars as Protocol Buffers

March 2026 - Present

Go, protobuf, gRPC, EBNF, Metaparsers, AST

gh: [accretional/proto-css](#)

- Can you imagine a universal type system that encodes all context-free grammars? So all you need to worry about is AST to AST transformations with the language-specific rules encoded as EBNF productions? That's what this is.
- Co-authored a **universal grammar metaparser** that generates the entire type system from the format's own grammar, so parsing and rendering are reflection over generated protobuf types with no per-format code to maintain or drift.
- Supporting 10+ formats (referenced here the project for the type system of CSS, the notoriously complicated language), then pushed to its limit by modeling the ARM64 instruction set from ARM's official spec, verified against real Apple Silicon down to the undocumented Apple AMX coprocessor.

REST APIs as Typed gRPC Services

Feb 2026 - May 2026

Go, Protocol Buffers, OpenAPI, Code Generation

gh: [accretional/openapi2proto](#)

- Every service usually has an OpenAPI spec defining their API contract, but service discovery would mean reading them manually, a huge pain if you want to build an API registry. What if you could discover it on demand via proto-reflection?
- Built a generator that produces a typed reflection-supported gRPC surface directly from an OpenAPI v3 spec (each RPC backed by the REST call) and regenerates from upstream.
- The hard part was surviving specs nobody hand-curated (unusual schema compositions, freeform objects, uncommon response codes, an unsupported spec version); proven by wrapping the full surface of the OpenAI (35 services, 242 RPCs) and Cloudflare (220 services) APIs.

OpenVINO Go Bindings

Nov 2025 - Jan 2026

Go, C++, CGO, OpenVINO

gh: [accretional/openvino-go](#)

- We needed fast inference on Intel CPUs, where OpenVINO beats ONNX Runtime, but no production-quality Go bindings for OpenVINO existed. Authored them from scratch as idiomatic Go/CGO over a three-layer architecture (public Go API, CGO bridge, C++ wrapper).
- Went past the reference implementation with async inference and variable-state management for recurrent models, benchmarking 40% lower latency than synchronous ONNX Runtime; 90%+ test coverage, now a production dependency behind the embedding and RAG services.

Local AI Toolkit for Apple Silicon

Oct 2025 - Dec 2025

Objective-C, Apple Vision, AVFoundation, Speech, Core Image

gh: [accretional/macros-vision](#)

Apple ships excellent on-device AI frameworks, but each needs its own glue code, and why rely on cloud services when you can do things on your own system? Unified Vision, Speech, AVFoundation, Sound Analysis, Natural Language, and Core Image into one CLI of Unix-composable subcommands (stdin/stdout), for private, low-latency inference with no server in the loop.

Publications

Machine Learning, Image Processing, Network Security and Data Sciences

2023

Prostate Cancer Grading Using Multistage Deep Neural Networks

Springer

Proposed three-stage ensemble deep learning method for automatic prostate cancer grading, achieving 92.38% classification accuracy on 10,000 histopathological images

Security in Computing and Communications (SSCC 2018)

2019

Network Anomaly Detection Using ANNs Optimised with PSO-DE Hybrid

Springer

Developed network anomaly detection system using ANNs optimized with hybrid PSO-DE algorithm, achieving 94% detection accuracy on network intrusion datasets

Skills

Programming Languages Python, Go, C++, CGO, TypeScript, Scala

Machine Learning & AI Inference & Systems OpenVINO, vLLM, LLM Inference, PyTorch, LoRA/PEFT, RAG, Model Optimization, Vector Search
Async Inference, Model Quantization, gRPC, Distributed Systems, Performance Profiling

Cloud & Infrastructure AWS, Docker, Modal, Google Cloud, CI/CD, Cloudflare, Envoy

Data & Storage SQLite, sqlite-vec, PostgreSQL, Redis, Vector Databases, NumPy/Pandas